

Hawaii Cropland Data Layers (HCDL) V2.0: A Satellite Embedding Approach to Crop Mapping

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Abstract— This study introduces Hawaii Cropland Data Layers (HCDL) V2.0, the first operational crop mapping product leveraging Google DeepMind's AlphaEarth Foundations Satellite Embedding dataset. The workflow replaces earlier multi-sensor image composites with 64-dimensional satellite embeddings to generate annual 10-meter resolution cropland maps for Hawaii in 2023 and 2024. The approach addresses challenges posed by persistent cloud cover, fragmented agricultural landscapes, and limited ground-reference data, while improving classification accuracy and consistency across 27 land-cover classes. Using Random Forest classifiers and a mixed sampling strategy, the embedding-based method achieved superior performance compared to previous composite-based models, with macro-average F1-score gains and reduced variability across classes. HCDL V2.0 provides a scalable, analysis-ready solution for agricultural monitoring, land-use planning, and disaster resilience in tropical regions. The datasets are publicly available and visualized through the AgriWatch web application, with HCDL 2025 currently under development.

Keywords— *Hawaii, Cropland Data Layers, Satellite Embedding dataset, Random Forest, Google Earth Engine, 10-meter resolution, AlphaEarth Foundations*

I. INTRODUCTION

Hawaii's agroecosystems are highly diverse, characterized by small farms, fragmented parcels, and varied crop types and management practices. Accurate, up-to-date agricultural land-cover information is essential for improving productivity, sustainability, ensuring food security, and resilience to extreme weather events and natural disasters. The Hawaii Agricultural Land Use Baseline (HALUB) [1], provided by the Hawaii Department of Agriculture, serves as a key crop-specific land-use dataset and offers valuable information for regional

assessments; however, its manual aerial image digitization process and five-year update cycle limit its ability to capture rapidly changing agricultural conditions. To address this gap, USDA's National Agricultural Statistics Service (NASS), in collaboration with the University of Hawaii at Manoa, initiated the development of annual 10-meter resolution Hawaii Cropland Data Layers (HCDL) [2][3].

Crop mapping in tropical regions such as Hawaii faces persistent challenges, particularly cloud cover, which obscures optical satellite imagery and complicates consistent data acquisition. In HCDL V1.x [3], this issue was mitigated using time-series 10-day median composites of Normalized Difference Vegetation Index (NDVI) and surface reflectance derived from Sentinel-Landsat multi-sensor data. These composites preserved crop phenology across the growing season but required extensive preprocessing and still struggled with fragmented cropping patterns and limited ground-reference data.

In September 2025, Google released the first annual global Satellite Embedding dataset, generated by DeepMind's AlphaEarth Foundations (AEF) model [4]. This dataset represents a major advancement in geospatial AI, integrating multiple Earth observation modalities—including optical, radar, LiDAR, and ancillary datasets—into a unified 64-dimensional embedding space. Each embedding vector encodes temporal trajectories and spatial context for every 10-meter pixel, capturing both spectral and semantic relationships across diverse landscapes. Unlike traditional spectral composites, embeddings are designed for generalization, enabling robust classification in regions with sparse labeled data [5].

For Hawaii, where persistent cloud cover and limited ground-reference samples have historically constrained crop mapping, this capability offers a transformative advantage. In this paper, we evaluate the effectiveness of AEF embeddings for crop-type classification and present a new operational workflow that replaces multi-sensor composites with embeddings to produce Hawaii Cropland Data Layers (HCDL) V2.0 for 2023 and 2024.

II. DATA

A. Input Data

This study utilized the AlphaEarth Foundations (AEF) Satellite Embedding dataset V1.0 (GOOGLE/SATELLITE_EMBEDDING/V1/ANNUAL) for cropland classification in Hawaii. At 10-meter spatial resolution, this dataset provides a 64-dimensional embedding vector that encodes temporal trajectories of surface conditions at and around each pixel, derived from multiple Earth observation sources. Although embeddings are less interpretable than conventional spectral inputs and vegetation indices, they effectively summarize relationships across multi-source, multi-modal observations. Figure 1 illustrates the embeddings for Hawaii in 2024, visualized using three axes of the embedding space as an RGB composite. The dataset is analysis-ready, eliminating pre-processing steps such as atmospheric correction, cloud masking, spectral transformations, and speckle filtering. This streamlined approach reduces data preparation complexity and improves classification performance.

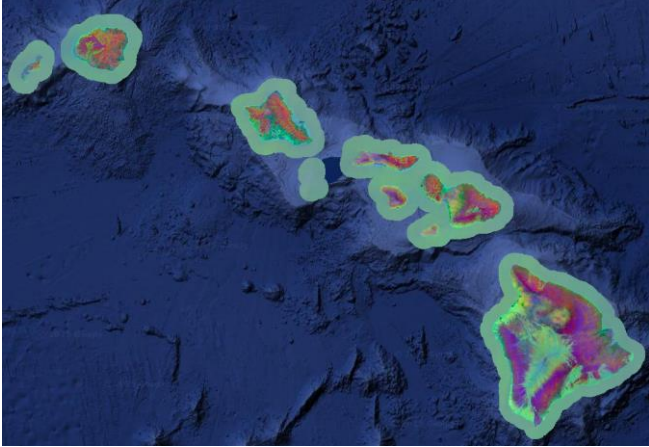


Fig. 1. Visualization of three axes of the embedding space as an RGB false color composite (R: “A01”, G: “A16”, B: “A09”).

B. Ground-reference Data

Ground-reference data were collected by the University of Hawaii at Manoa through fieldwork conducted between December 2023 and October 2025 across Kauai, Oahu, Maui, Molokai, Lanai, and Hawaii (Big Island). Additional interpretation of satellite and aerial imagery was performed to label crop-specific land cover, using sources such as USDA NAIP 2022 60 cm CIR imagery, high-resolution Google Earth imagery, and PlanetScope time-series imagery from 2023 and 2024. These datasets, combined with the HALUB 2020 dataset, facilitated identification of active agricultural areas and

exclusion of fallow lands. To improve differentiation between agricultural and non-agricultural classes, the 2021 1-meter resolution Hawaii Tree Canopy GIS Layer [6] was incorporated to refine training labels, particularly for distinguishing tree canopy from cropland and delineating “commercial forest” and fruit tree species polygons.

III. METHODOLOGY

A. Processing Units

Localized processing often yields more accurate crop mapping results than statewide approaches [7]. In Hawaii, similar cropping patterns typically occur within the same watershed. The Watershed Boundary Dataset (WBD), developed by the USGS, provides hydrologic units (HUs) delineated strictly by hydrologic principles. At the HUC10 level, the major Hawaiian Islands are divided into 52 watersheds. Given the complexity of cropping practices and the highly fragmented nature of parcels, this study performed classification independently for each watershed rather than as a single statewide unit.

B. Sampling Strategies

Sampling strategy significantly influences machine learning classification performance. Random Sampling (RS), while common, tends to over-represent dominant classes and under-represent smaller land-cover types. Stratified Random Sampling (SRS) improves representation but can disproportionately amplify small-area classes, leading to over-classification [8]. This issue is pronounced in Hawaii’s fragmented cropland patterns. To balance representation, we employed a mixed strategy combining RS and SRS: within each watershed, 20,000 samples were drawn randomly, supplemented by 1,000 stratified samples per land-cover type. This approach produced a more balanced and robust training dataset.

C. Classification Algorithms

HCDL aims to map major, specialty, and culturally significant crops grown in Hawaii, including coffee, macadamia nuts, banana, pineapple, citrus, taro, plumeria, dracaena, papaya, avocado, coconut, sugarcane, sunflower, sweet potatoes, sweet basil, seed crops (primarily corn), and commercial forests. Due to limited ground-reference data, certain tropical fruit species (e.g., mango, litchi, longan, cacao, dragon fruit, jackfruit, breadfruit, and rambutan) were grouped under “Other Exotic Fruits,” while crops such as tomato, pumpkin, turmeric, mustard, eggplant, watermelon, and onion were grouped under “Misc Veggies & Fruits.”

Previous studies compared Random Forest (RF) [9] and Convolutional Neural Networks (CNN) [10][11] for crop mapping in Hawaii. While CNNs excelled in segmentation for some vegetation types, RF achieved higher accuracy and faster processing when classifying multi-sensor composites [3]. Satellite embeddings, being unit-length vectors requiring no normalization, are well-suited for tree-based classifiers [4]. Therefore, this study applied RF to classify AEF’s 64-dimensional embeddings using the `smileRandomForest` implementation in Google Earth Engine, configured with 100 trees and a `bagFraction` of 0.5.

D. Model Training and Validation

Using the mixed sampling strategy, datasets were generated for each watershed and split into 80% training and 20% validation. RF classifiers were trained per watershed and applied for crop-type classification. Accuracy was assessed using an independent testing dataset of 11,681 samples (10,000 random and 1,681 stratified). Metrics included Overall Accuracy (OA) and per-class F1-score (F1). Due to limited ground-reference data in 2023, model training and validation were conducted only for HCDL 2024. The 2023 layer was derived by applying 2024-trained models to 2023 embeddings.

IV. RESULTS AND DISCUSSION

A. Accuracy Assessment

Figure 2 illustrates the satellite embedding-based, 10-meter resolution HCDL 2024 containing 21 agricultural and 6 non-agricultural categories generated using Google Earth Engine (GEE) [12]. Figure 3a shows comparison of per-class F1 scores for 27 categories using classification derived from two datasets: 10-day Multi-Sensor Image Composites and AEF Satellite Embeddings (validation set: 11,681 samples). Results indicate that the satellite embedding-based crop mapping generally outperforms the 10-day composite approach, with a macro-average improvement from 0.955 to 0.965. Gains are most pronounced for heterogeneous crops and developed areas (e.g., Coffee, Macadamia, Other Exotic Fruits, Other Crops, Developed), while classes with near-perfect scores (e.g., Open Water, Citrus, Sugarcane) remain unchanged. Minor regressions occur in a few categories, notably Corn (-0.023), but overall trends indicate a robust advantage for AEF embeddings in multi-class land cover mapping.

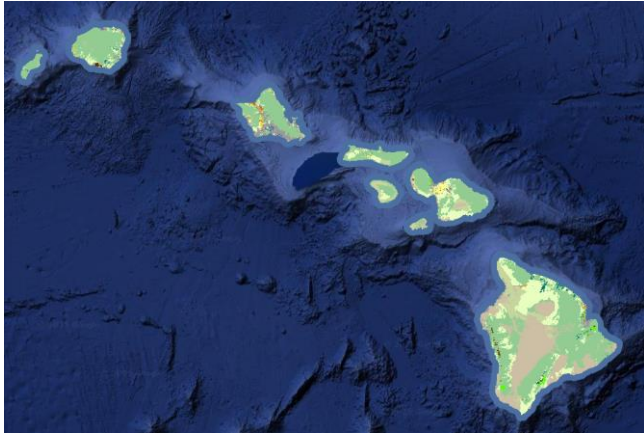


Fig. 2. 2024 10-meter resolution Hawaii Cropland Data Layer (HCDL).

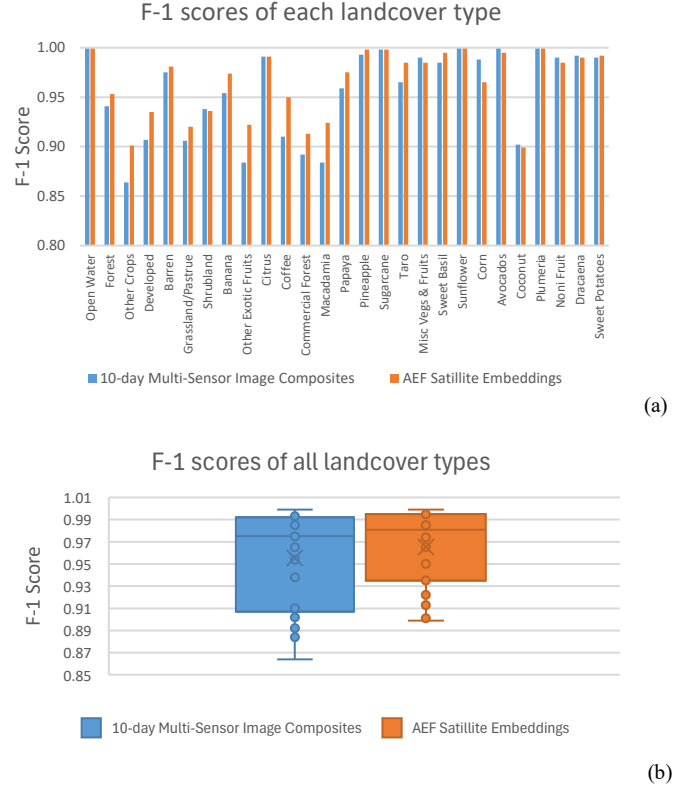


Fig. 3. Comparison of per-class F1 scores for 27 land cover and crop categories.

Figure 3b summarizes the distribution of per class F1 scores. Relative to the 10 day composites, AEF embeddings exhibit a higher median (0.981 vs. 0.975) and a smaller interquartile range (IQR) (0.058 vs. 0.083), indicating improved typical performance and reduced variability across classes. The lower whisker increases (min: 0.899 vs. 0.864), and no outliers are present under the $1.5 \times \text{IQR}$ rule, suggesting broad based gains rather than isolated improvements. Consistent with these trends, the share of classes with $F1 \geq 0.95$ rises from 63.0% to 70.4%, while the number at $F1 \leq 0.90$ drops from 4 to 1, reinforcing the conclusion that AEF delivers both higher accuracy and greater consistency across the validation set.

B. Landcover Change Detection

The 2023–2024 HCDL layers derived from AEF satellite embeddings reveal significant land-cover dynamics across the Hawaiian Islands, highlighting both agricultural expansion and stability:

- **Kauai:** The Kaumakani area (Figure 4a) exhibits a clear shift from fallow lands to a diversified agricultural mosaic. The 2024 HCDL shows substantial expansion of productive acreage driven by new energy and food crops. Previously uncultivated parcels have been converted into sunflower fields for biodiesel and intensive row crops such as watermelon and mixed vegetables. This “fallow-to-productive” signal underscores the region’s evolving role in local sustainability and food security initiatives.

- Oahu: Agricultural footprints remain relatively stable across the observation period, demonstrating resilience despite intensifying competition for land resources. However, this stability coexists with visible encroachment from urbanization and renewable energy infrastructure, particularly in the Central and Ewa plains (Figure 4b left). The data highlights a spatial tension between persistent cultivation and the expansion of competing land uses, signaling potential pressure on key agricultural corridors.
- Maui: In the Central Valley (Figure 4c), the 2023–2024 HCDL layers capture a critical maturation phase. Unlike the clearing and planting signatures of previous years, recent data shows distinct canopy closure across approximately 13,000 acres of permanent orchards, primarily citrus. This marks a stable transition from former sugarcane monoculture to a diversified agricultural mosaic, where established tree crops are interspersed with rotational fields of watermelon and onion.
- Molokai: The Ho’olehua plains (Figure 4d) exhibit a dual-anchor of agricultural stability. The seed corn area maintains a high-intensity footprint, with spectral signatures reflecting dense planting of short-stature corn varieties. Adjacent to these seasonal field crops, the papaya orchards display a resilient, permanent “Tropical Fruit” signature. Together, these areas form the island’s economic core—a mix of industrial seed production and premium export-fruit cultivation that has successfully maintained active acreage despite regional drought pressures.
- Hawaii Island: The Hilo and Hamakua agricultural corridors (Figure 4b right) present a distinct “dual-system” landscape where permanent tree canopies coexist with dynamic, open-field farming. Established orchards such as macadamia and lychee provide a stable perennial signature, while large tracts are frequently plowed and rotated for high-value root crops including ginger, turmeric, and sweet potato. This creates a complex visual mosaic where rapid changes—from green vegetation to bare soil—represent active crop rotation rather than development or abandonment.

V. SUMMARY AND FUTURE WORK

This study demonstrates the successful integration of Google DeepMind’s AlphaEarth Foundations Satellite Embedding dataset into operational crop mapping for Hawaii, resulting in the first 10-meter resolution Hawaii Cropland Data Layers (HCDL) V2.0 for 2023 and 2024. By replacing earlier multi-sensor image composites with satellite embeddings, the new workflow significantly improves classification accuracy and consistency across diverse crop types, particularly in cloud-prone and fragmented agricultural landscapes. The embedding-based approach streamlines data preparation, reduces reliance on extensive ground-reference datasets, and offers a scalable solution for annual cropland monitoring. These advancements not only enhance agricultural decision-making and land-use

planning in Hawaii but also provide a blueprint for advancing USDA-NASS’s 10-meter resolution nationwide Cropland Data Layers [7] production especially in data-limited regions.

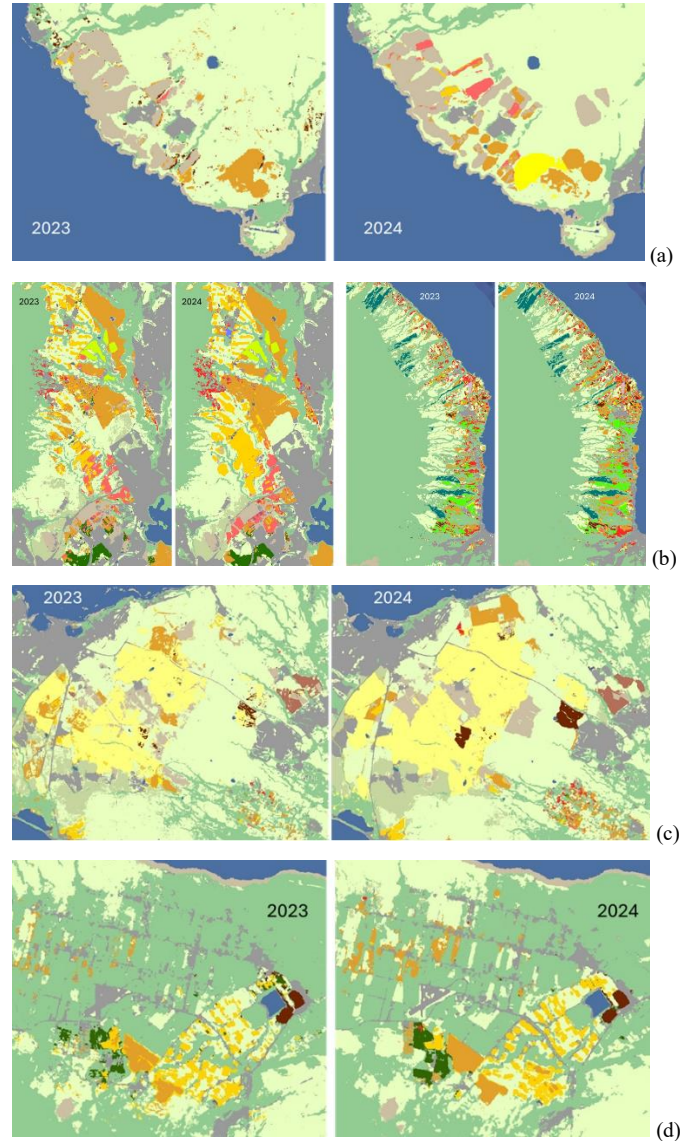


Fig. 4. Land-cover change and dynamics across the Hawaiian Islands

USDA-NASS has officially released HCDL V2.0 for 2023 and 2024 [13], which can be downloaded from the official website and visualized via the Google Earth Engine web application “AgriWatch” [14]. Development of HCDL 2025 using the same methodology is underway, with release expected in spring 2026.

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